

Problem 5

Two balls of equal mass, $m_1 = m_2 = 2$ g, move with equal speeds, $v_1 = v_2 = 20$ cm/s, along paths that are perpendicular to each other. What is the total momentum of both balls?

Solution

The momentum of a body is given by the formula:

$$\vec{p} = m\vec{v}.$$

For ball 1:

$$\vec{p}_1 = (2 \times 10^{-3}) \times (0.2) = 4 \times 10^{-4} \text{ kg} \cdot \text{m/s}.$$

For ball 2:

$$\vec{p}_2 = (2 \times 10^{-3}) \times (0.2) = 4 \times 10^{-4} \text{ kg} \cdot \text{m/s}.$$

Since the velocities are perpendicular, the total momentum is the vector sum:

$$\vec{p}_{\text{total}} = \vec{p}_1 + \vec{p}_2.$$

The magnitude of the total momentum is calculated using the Pythagorean theorem:

$$p_{\text{total}} = \sqrt{p_1^2 + p_2^2}.$$

Substitute the values:

$$p_{\text{total}} = \sqrt{(4 \times 10^{-4})^2 + (4 \times 10^{-4})^2}.$$

$$p_{\text{total}} = \sqrt{16 \times 10^{-8} + 16 \times 10^{-8}}.$$

$$p_{\text{total}} = \sqrt{32 \times 10^{-8}}.$$

$$p_{\text{total}} = 5.66 \times 10^{-4} \text{ kg} \cdot \text{m/s}.$$

Final Answer:

The total momentum of both balls is approximately:

$$5.66 \times 10^{-4} \text{ kg} \cdot \text{m/s}.$$